Pattern Matching with Sequences

Example 1 : Empty Sequence Pattern

Pattern matching on the empty sequence:

$$
\begin{array}{|l}
f : \mathrm{seq}\ \mathbb{N} \longrightarrow \mathbb{N} \\
\hline
f(\langle\rangle) = 0
\end{array}
$$

Example 2 : Cons Pattern (Head land Tail)

Decomposing a sequence into head and tail:

$$
\begin{array}{|l}
g : \mathrm{seq}\ \mathbb{N} \longrightarrow \mathbb{N} \\
\hline
g(\langle\rangle) = 0 \\
\forall\, x : \mathbb{N};\ s : \mathrm{seq}\ \mathbb{N} \bullet g(\langle x \rangle \frown s) = x
\end{array}
$$

Example 3 : Recursive Sum

Computing the sum of elements using pattern matching:

$$
\begin{array}{|l}
total : \mathrm{seq}\ \mathbb{N} \longrightarrow \mathbb{N} \\
\hline
total(\langle\rangle) = 0 \\
\forall\, x : \mathbb{N};\ s : \mathrm{seq}\ \mathbb{N} \bullet total(\langle x \rangle \frown s) = x + total(s)
\end{array}
$$

Example 4 : Cumulative Total

A more descriptive example from the solutions:

$$
\begin{array}{|l}
cumulative\_total : \mathrm{seq}\ \mathbb{N} \longrightarrow \mathbb{N} \\
\hline
cumulative\_total(\langle\rangle) = 0 \\
\forall\, x : \mathbb{N};\ s : \mathrm{seq}\ \mathbb{N} \bullet \\
\qquad cumulative\_total(\langle x \rangle \frown s) = x + cumulative\_total(s)
\end{array}
$$

Example 5 : Filter Pattern

Filtering sequences based on a condition:

$$
\begin{array}{|l}
positives : \mathrm{seq}\ \mathbb{Z} \longrightarrow \mathrm{seq}\ \mathbb{Z} \\
\hline
positives(\langle\rangle) = \langle\rangle \\
\forall\, x : \mathbb{Z};\ s : \mathrm{seq}\ \mathbb{Z} \bullet \\
\qquad positives(\langle x \rangle \frown s) = \mathsf{if}\ x > 0 \\
\qquad \mathsf{then} \langle x \rangle \frown positives(s) \\
\qquad \mathsf{else} positives(s)
\end{array}
$$